Computer Networks : Assignment 1

1. Find a situation in which you may use Wireshark to observe or track protocols as they are being used while engaging with a network application of your choice. Write a brief report that details, step by step, how Wireshark is being used to examine or follow protocols.

Wireshark is network packet analysis tool. It helps to understand and view the internals of network packet workflow and display them at a microscopic level. In summary, it is software which shows what's happening inside the various protocols of the network stack.

Usage of the wireshark:
Generally, System & Network Admins, Security engineers, QA teams use the wireshark for investigating, troubleshooting and testing the network based applications and related security & network problems.

Scenario:

In day-to-day life, we use broswer to find out various content online. What happens when we visit any website say "http://www.testingmcafeesites.com/testreputation_highrisk.html" in the address bar of the browser? Wireshark can be used to investigate the HTTP traffic between the browser and the web server. Let's see in details.

Lab Setup details:

- Operating System : Kali Linux
- Softwares: Wireshark, Firefox, Terminal
- Host IP address: 172.16.1.6

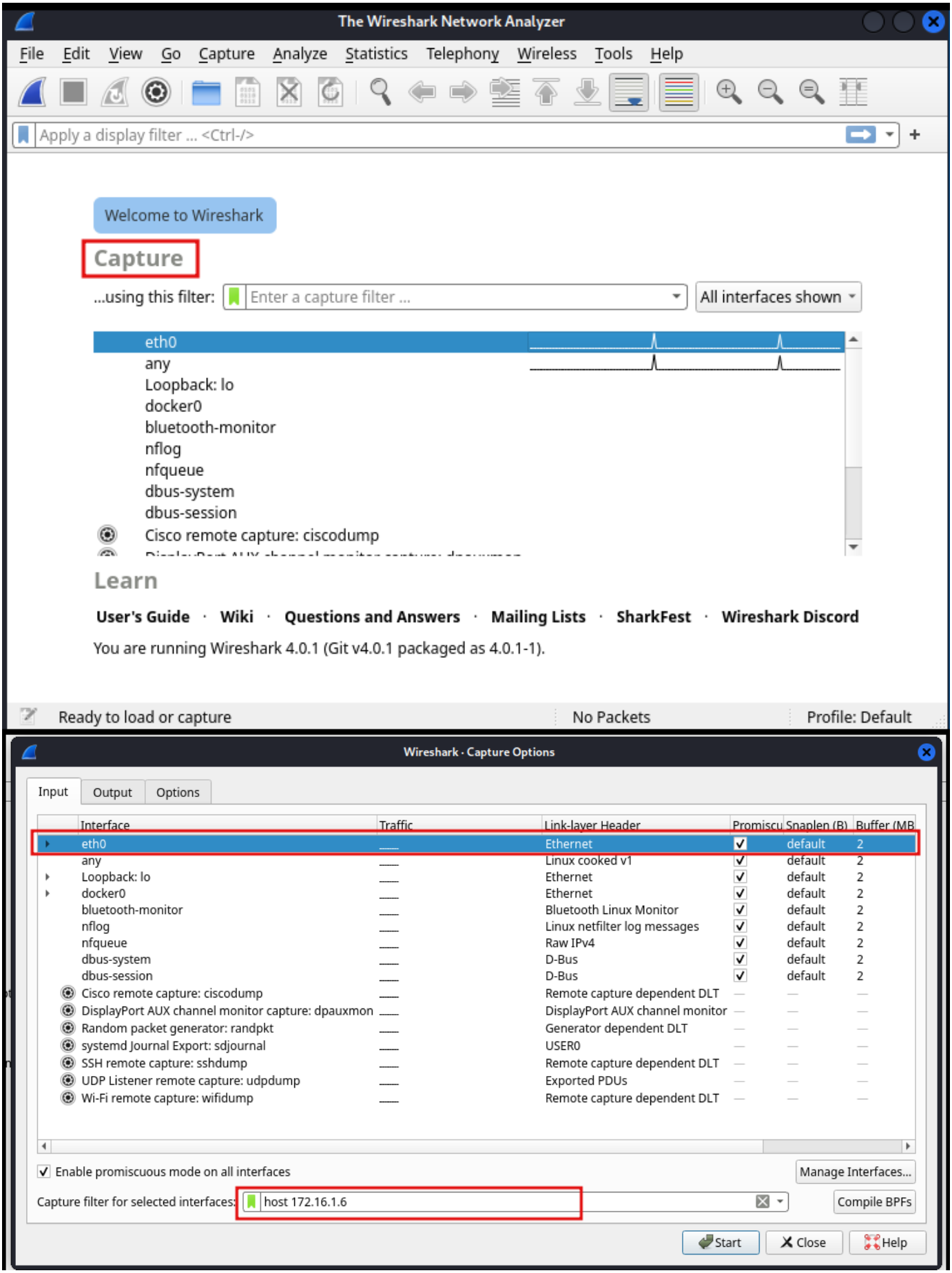
```
┌──(root㉿kali)-[/home/kali]
└─# ifconfig eth0
```

Brief Report:

Step 1: Start the Wireshark

```
┌──(root㉿kali)-[/home/kali]
└─# wireshark &
```

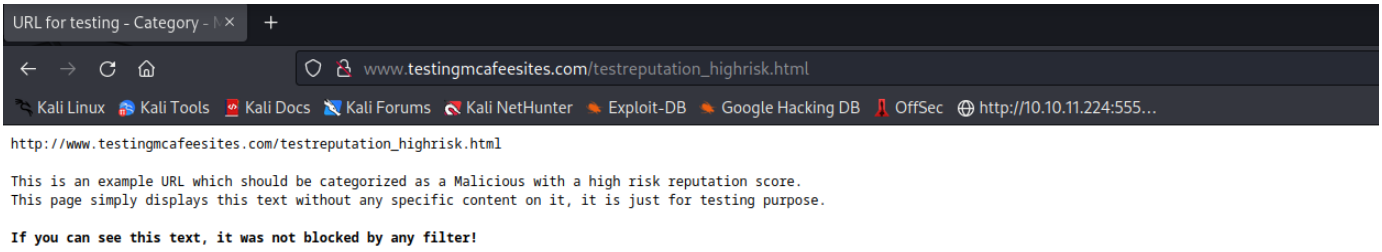
Note: Run Wireshark as root, which is required for capturing raw packets.
Step 2: Select Network Interfaces on the Wireshark GUI and apply the capture filter as "host 172.16.1.6"



Step 3: Open the web browser - firefox

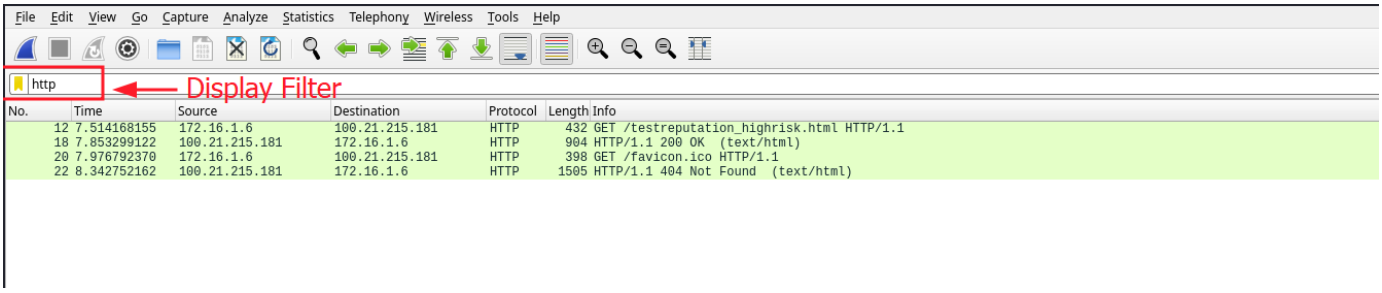
```
┌──(kali㉿kali)-[~]
└─$ firefox &
```

Step 4: Type "http://www.testingmcafeesites.com/testreputation_highrisk.html" in the address bar



Step 5: Stop capturing packets after 1 minutes.

Step 6: Apply HTTP display filter

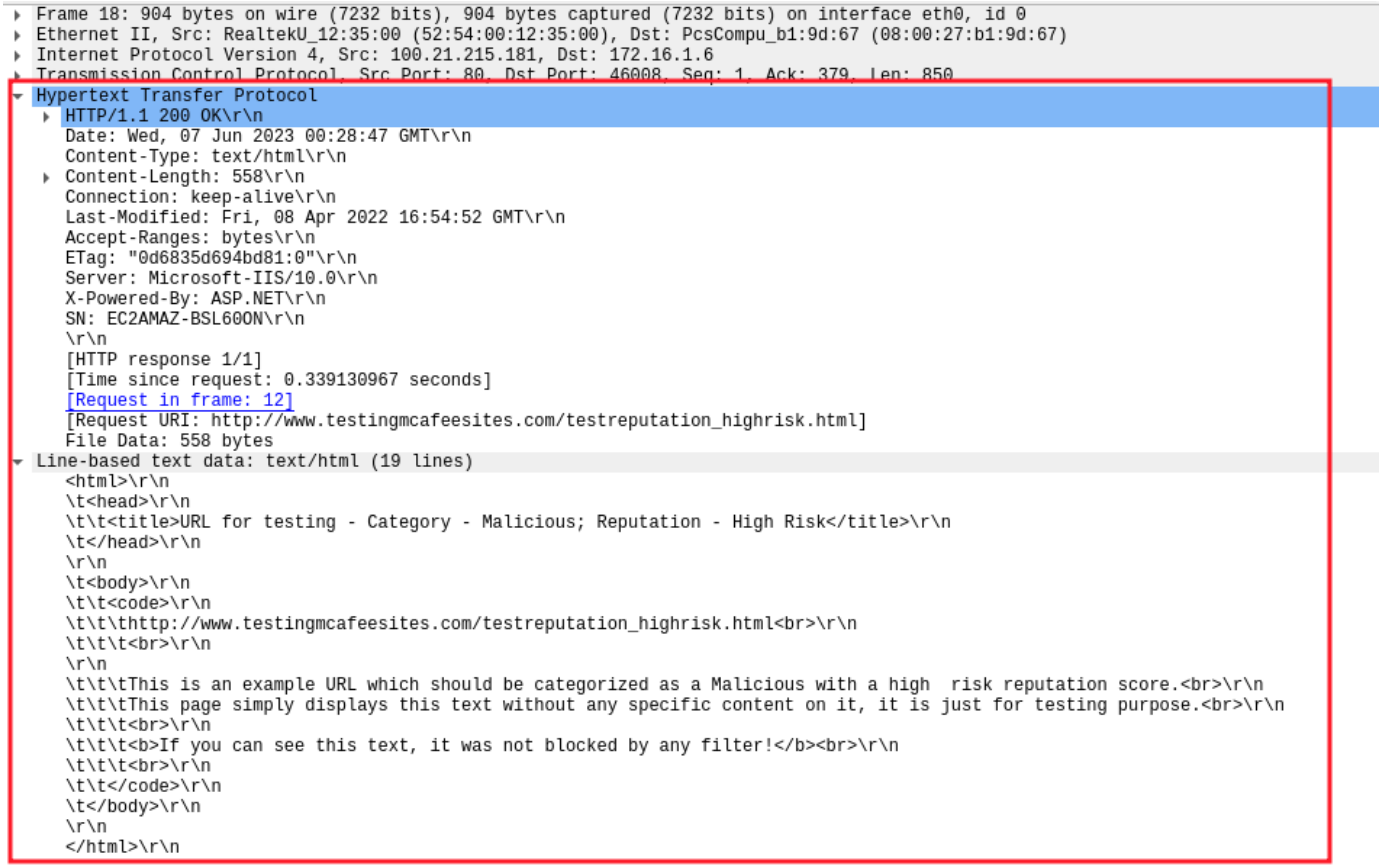


Step 7: HTTP trafic anlysis

Packet 12 is HTTP request packet which is sent from host (browser) to server (100.21.215.181). Click on the packet 12 to see the content of the request. It consists of various information such as HTTP request method, Host, User-Agent etc.

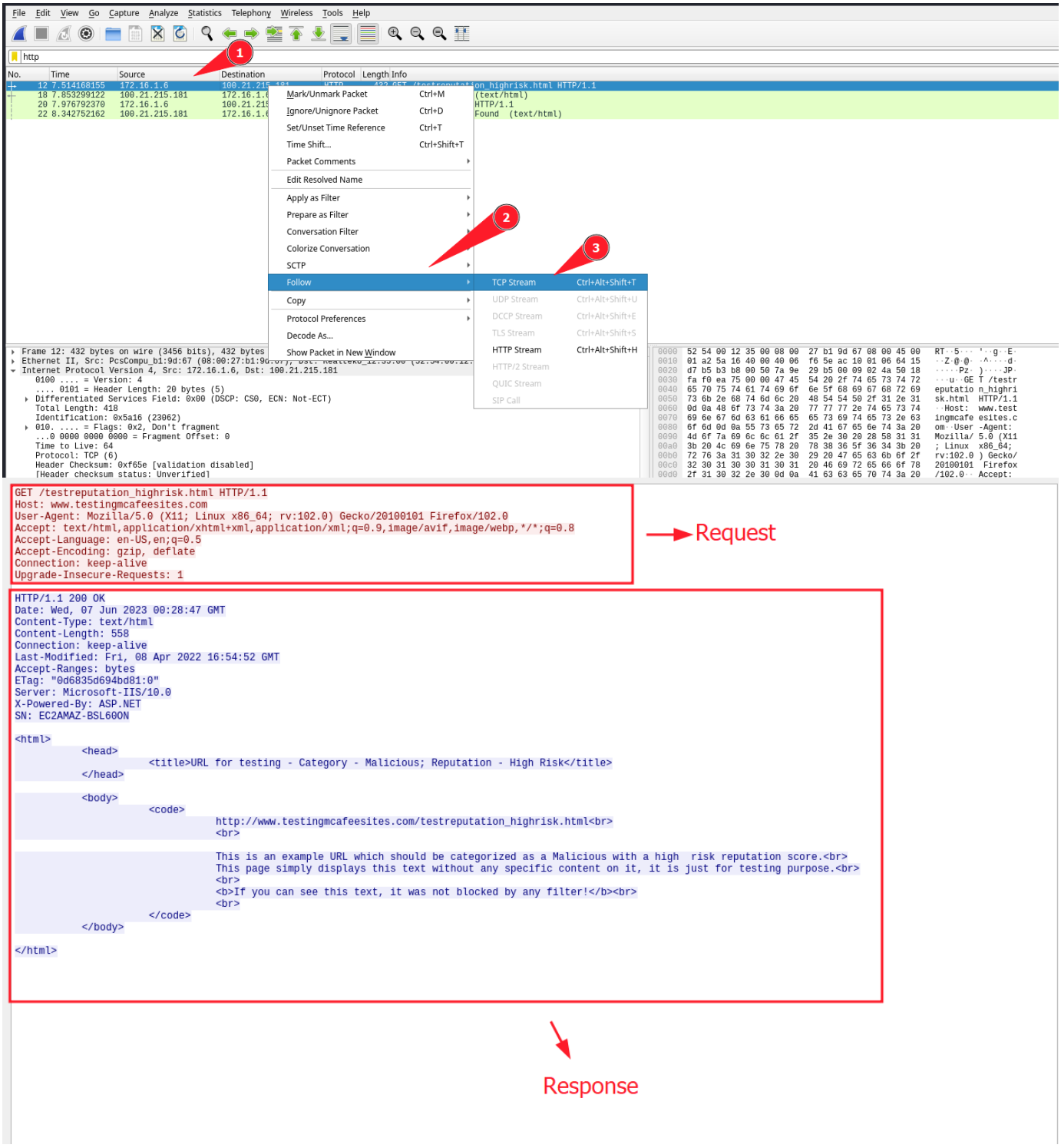


Packet 18 is the HTTP response packet which is received from server to host. Click on the packet 18 to see the content of the request. It consists of various information such as HTTP version, Date, Content-type, Content-Length, Server etc. along with message body



HTTP response packet

HTTP request-reponse can be view in single windows with much human-readable format. Right-Click on the packet 12 and select follow -> TCP Stream



There is much more happening at TCP level since we focused on HTTP protocol, request and response packets analysis is performed. In this way we can analyse the HTTP protocol using the wireshark.

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2. Briefly explain UDP network protocol. Demonstrate a comparison (similarity/dissimilarity if any) of the protocol as learned from the textbook with the Wireshark output.

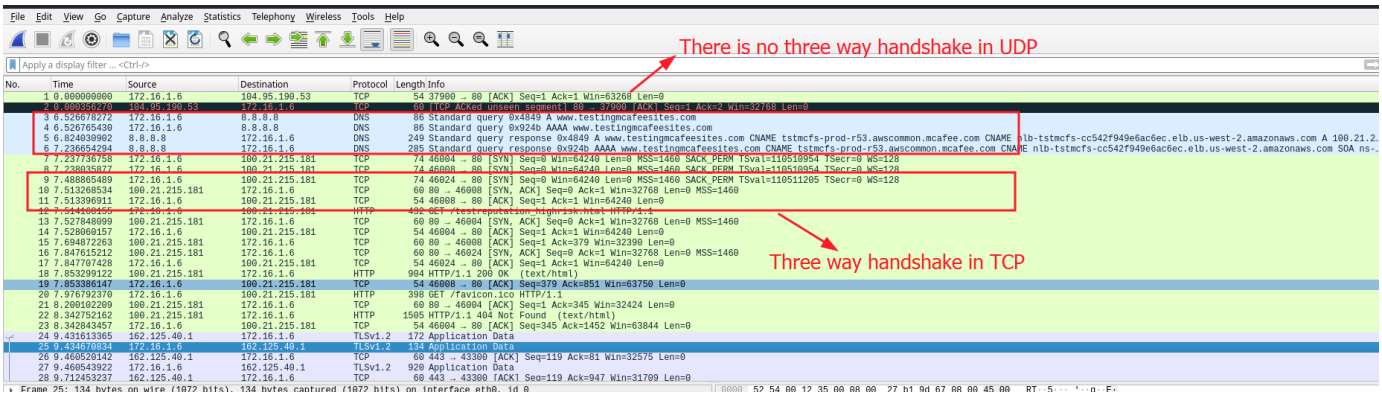
UDP (User Datagram Protocol) is a simple and connectionless network protocol. It is often used for time-sensitive transmission where speed is crucial such as video buffering or DNS lookup. DNS is a application layer protocol which uses UDP. Connection-less, low overhead, Unreliability, No congestion control are key characteristics of UDP protocol.

Packet Capturing procedure:

1. Start the Wireshark
2. Select Network Interfaces on the Wireshark GUI and apply the capture filter as "host 172.16.1.6"
3. Open the web browser - firefox and type "http://www.testingmcafeesites.com/testreputation_highrisk.html" in the address bar.
4. Stop capturing packets after gathering sufficient packets.

We learned that UDP is connection-less i.e. it is fire and forget. Once the packet is sent, sender does not wait for the acknownlegment packet. We can observe that no three way handshake is performed in the DNS resolution unlike TCP protocol. In the packet list pane of wireshark, we can observer that packet 2 and packet 3 are DNS query request packet with type A and AAAA record respectively. Similarly, packet 5 & packet 6 are corresponding query response packets which contains the answer for the asked query.

Three way handshake for TCP:
Packet 9 is the syn packet, packet 10 is the syn/ack packet and packet 11 is the ack packet.



In this way, a comparision between UDP and TCP protocol based on a connection-less property is demonstrated.

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3. Using Wireshark to demonstrate TCP three-way handshake.

Reliabilty is one of the characteristic of TCP protocol. Three-way handshake is mechanisms which ensures connection between client and server. Unlike the connection-less protocol like UDP. In TCP three way handshake, two communiationg parties are involved. Let's say device C and device S. C wants to communicate with S. C starts the communication by sending a packet then S recieves the packet. S acknowledge the connection started by C by sending packets to C. Finally C sends the ackowledgement Thus completing the three steps involved. In this steps various parameters and information is exchanged to ensure the reliability, error-checking and congestion control.

Packet Capturing Procedure:

1. Start the wireshark with high privilege
2. Select the network interface on the wireshark GUI with required capture filter as "host IP_Address" which starts capturing the packets.

3. Open the web browser such as firefox or chrome, and type "http://www.testingmcafeesites.com/testreputation_highrisk.html" in the address bar
4. Stop capturing packets after getting sufficient packets i.e. capture packets at least for 1 min.
5. Apply TCP as diplay filter.

Steps involved in TCP three-way handshake:

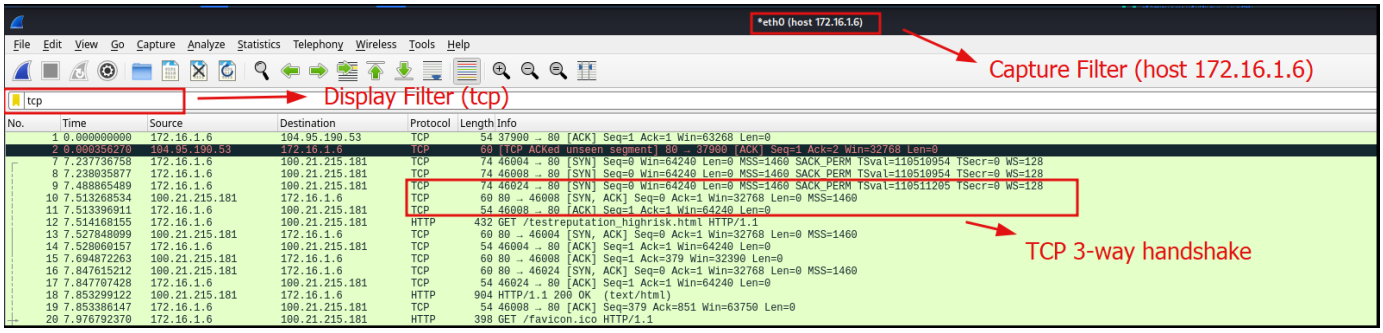
Step 1: Initiation of the connection from the 1st side.
Let's say devic C wants the service hosted by the devic S. It is inferred that S has open ports and ready to accept the new connection since it is hosting the service. In the this step, device C say, "Hey S, I want to start the connection to get the service". In networking world, it is done using sending special packet called SYN packet over the IP network. This data packet is labelled by using random number so that device S knows it, also it is helpful to send the next packet in order.

Step 2: Acknowlegement of the connection initiated and starting the connection from other end.
Once the SYN packet send by device C is recieved by device S. Now it's S turns to say, "Hey C I have recieved your message. Now I want to start the connection my end." In the computer networking langauge, device S sends the SYN/ACK packet to device C which serve the purpose of both, starting the connection and acknowledgement of the message sent by device C. Here labelling the packets involes two numbers. SYN contains random number on the device S side where ACK contains the SYN number sent by device C plus 1.

Step 3: Final acknowlegement of the connection from the 1st side.
Till now both the parties exchange the message once. How come device S knows that packet sent by it is received by device C. Therefore, device C send the acknowledgement of message sent by device S. In other words, C says "Hey, S I have received your message. Now you are good to allow me to enjoy the service". Labelling the packets is necessary in order to receive the packets in sequence.

In summary,

- Send the sync packet (packet no: 9 → source to the destination i.e. 172.16.1.6 to 100.21.215.181)
- Send the sync + acknowledgment packet (packet no: 10 → destination to source i.e. 100.21.215.181 to 172.16.1.6)
- Final acknowledgment (packet no. 11 → source to destination 172.16.1.6 to 100.21.215.181)

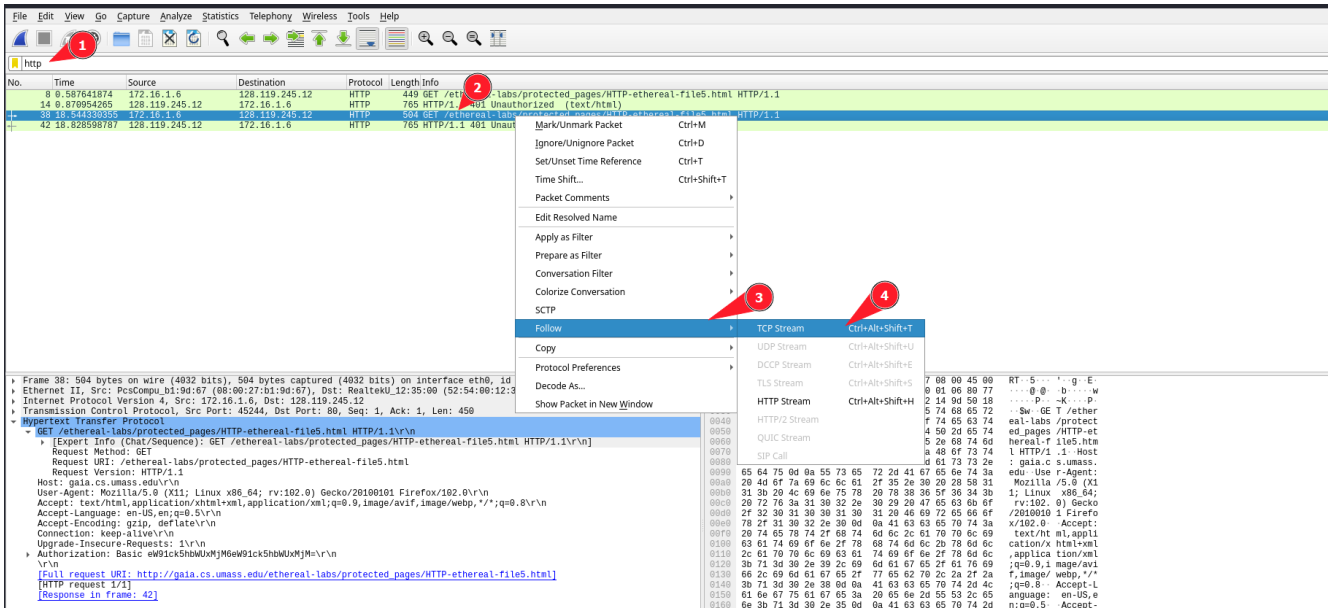


4. Go to Type 'yourName123' as both username and password and submit [replace yourName with your last name]. This is a website where the password is unencrypted. Provide a screen-shot of WireShark showing the unencrypted password.

Packet Capturing procedure:

1. Start capturing the packets using wireshark.
2. Once wireshark is started, visit the url "http://gaia.cs.umass.edu/ethereal-labs/protected_pages/HTTP-ethereal-file5.html" using web browser such as firefox.
3. Stop capturing packets after gathering enough packets.
4. Get the unencrypted password

   Apply HTTP as display filter, Right Click on the HTTP packet no 38 and select follow -> TCP stream.



   In the authentication header, we can see the authorisation which contains the credentials to authenticate a user-agent with a server.

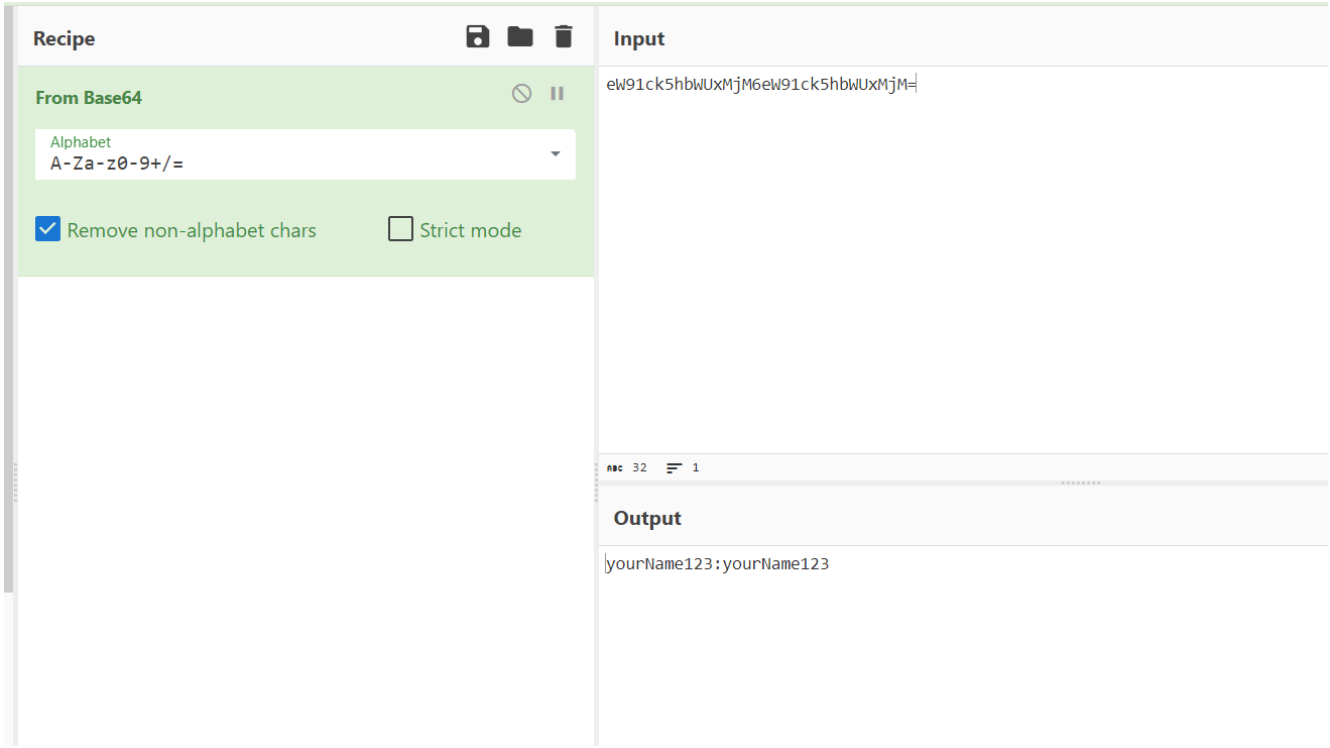
```
GET /ethereal-labs/protected_pages/HTTP-ethereal-file5.html HTTP/1.1
Host: gaia.cs.umass.edu
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:102.0) Gecko/20100101 Firefox/102.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Authorization: Basic eW91ck5hbWUxMjM6eW91ck5hbWUxMjM=

HTTP/1.1 401 Unauthorized
Date: Sat, 22 Jul 2023 03:05:02 GMT
Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3
WWW-Authenticate: Basic realm="eth-students only"
Content-Length: 381
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested.  Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
</body></html>
```

```
Authorization: Basic eW91ck5hbWUxMjM6eW91ck5hbWUxMjM=
```

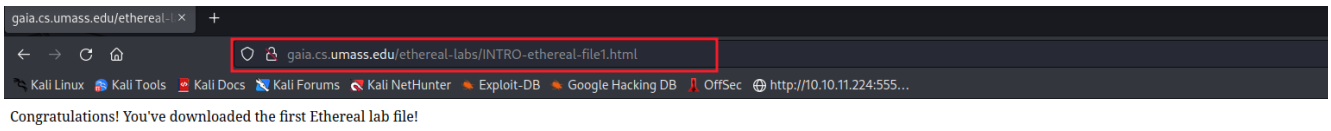
Here credentials are base64 encoded which can easily seen by decoding it.



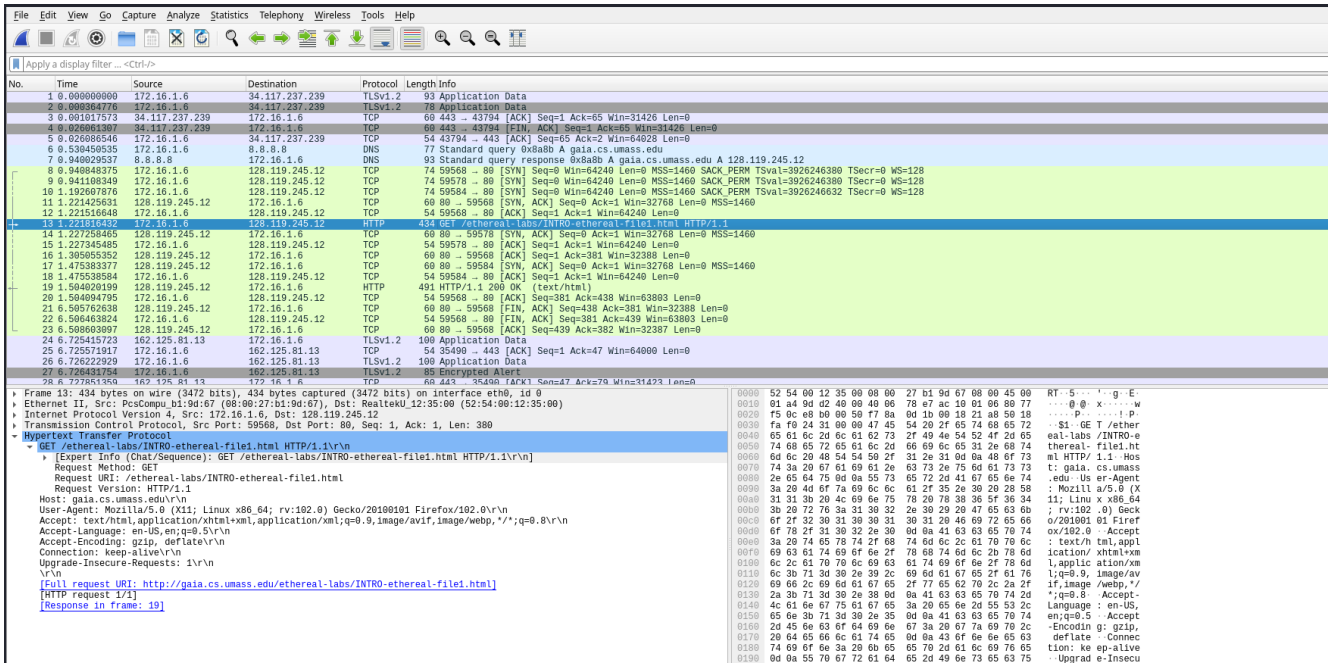
5. Click the following URL. To load the above page, what version of HTTP is the server using? Provide screenshot using for both the browser and Wireshark.

Packet Capturing procedure:

1. Start capturing the packets using wireshark.
2. Once wireshark is started, visit the url "http://gaia.cs.umass.edu/ethereal-labs/INTRO-ethereal-file1.html" using web browser such as firefox.

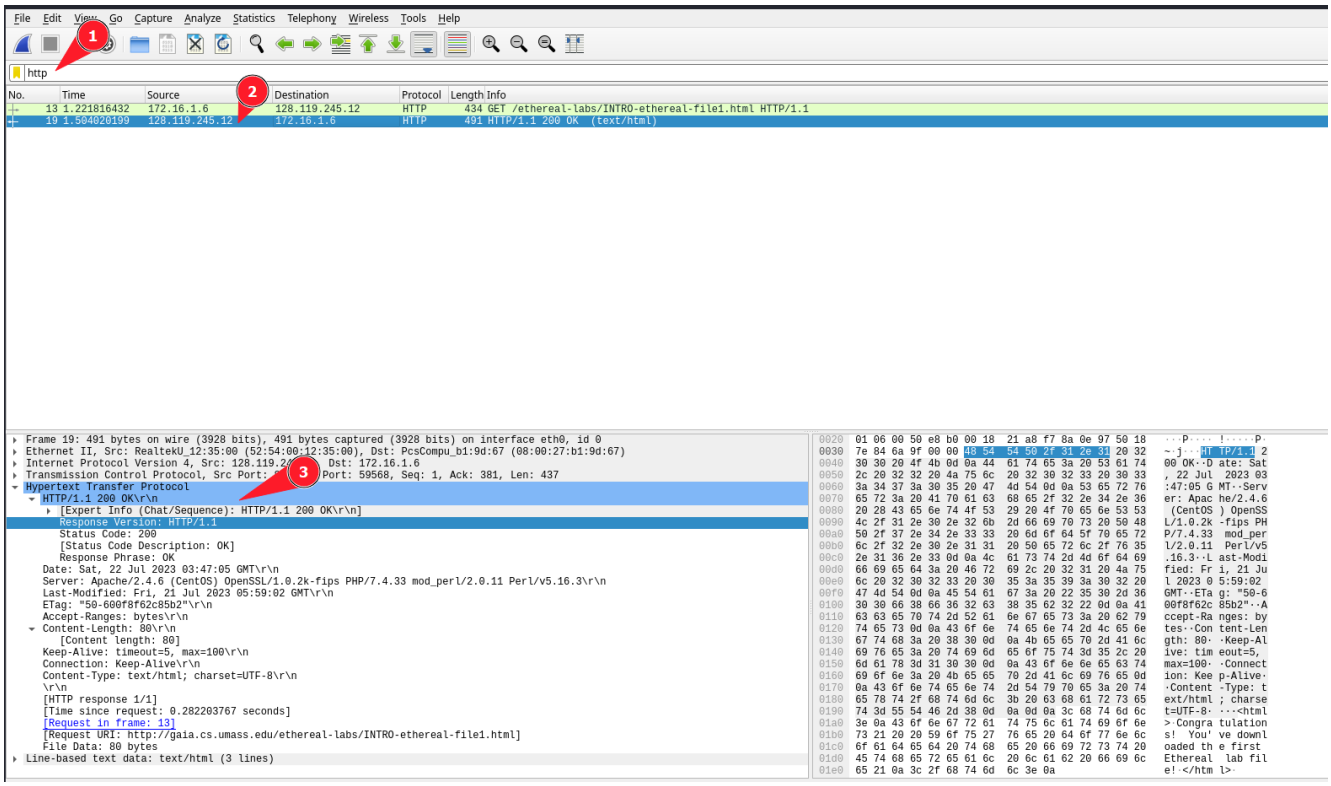


3. Stop capturing packets after gathering enough packets.



4. Get the HTTP version used by the server

   Apply HTTP as display filter, click on the HTTP response packet.



   We can the in the Packet Details pane of wireshark, server is using HTTP 1.1 version

```
HTTP/1.1
```

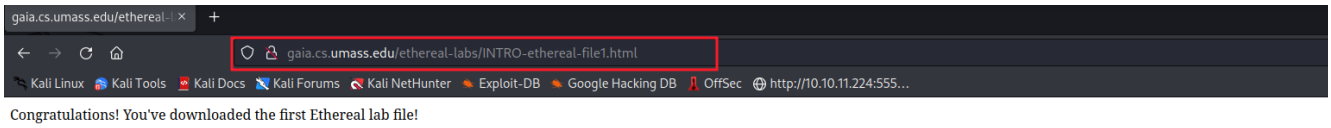
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6. What is the Internet address of the gaia.cs.umass.edu? What is the Internet address of your computer?

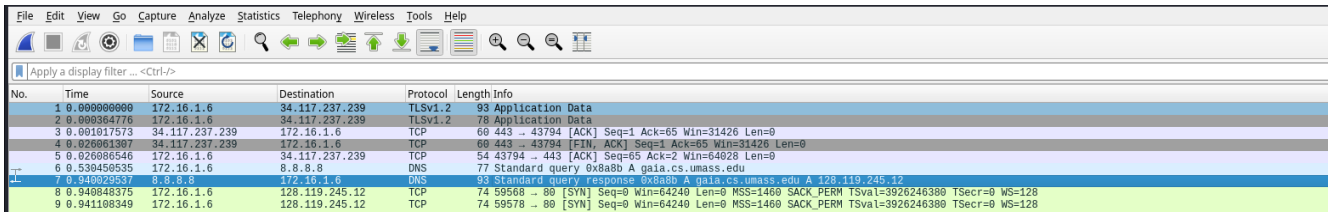
Packet Capturing procedure:

1. Start capturing the packets using wireshark.

2. Once wireshark is started, visit the url "http://gaia.cs.umass.edu/ethereal-labs/INTRO-ethereal-file1.html" using web browser such as firefox.

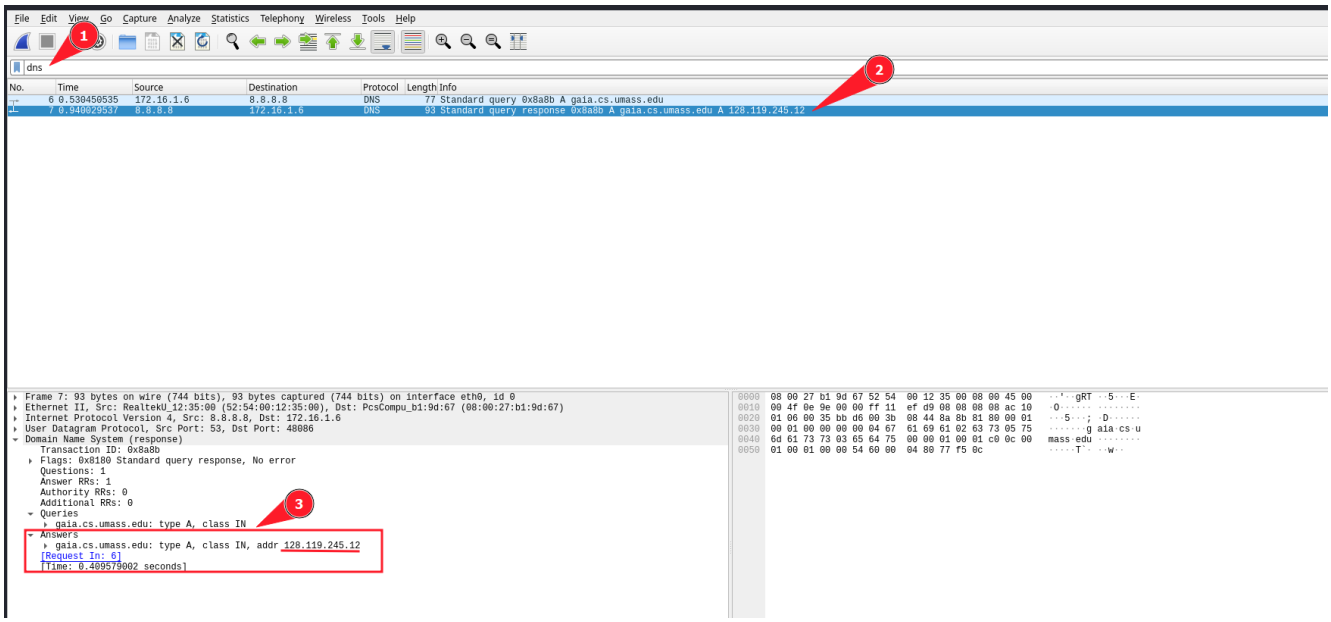


3. Stop capturing packets after gathering enough packets.



4. Obtain the Internet Protocol address of gaia.cs.umass.edu.

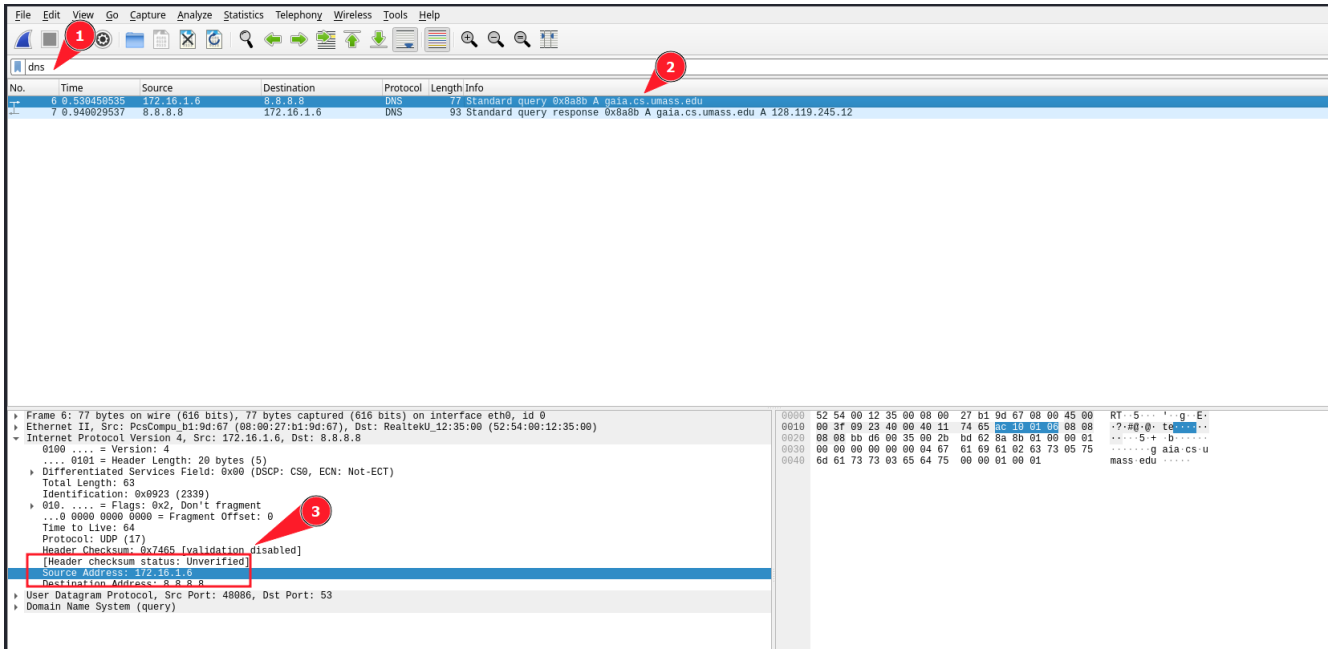
   Apply DNS as display filter, click on the query response packet, expand queries and answer tabs of the dns packet



   We can see in Packet Details pane of wireshark, IP address of the server: gaia.cs.umass.edu is `128.119.245.12`

5. Obtain the Internet Protocol address of my computer.

Apply DNS as display filter, click on the query packet



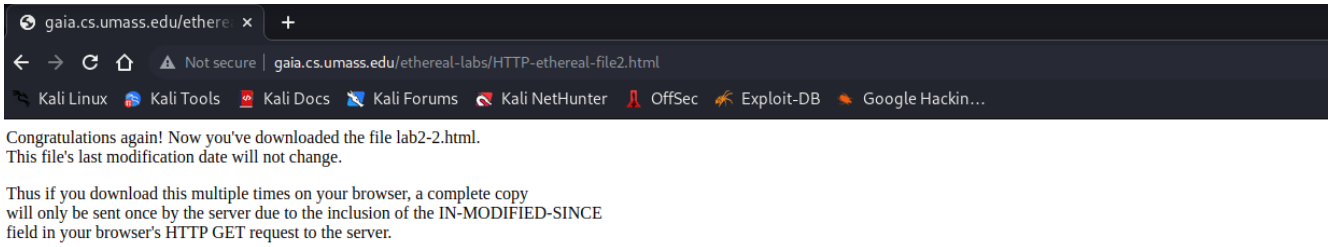
We can see in Packet Details pane of wireshark, IP address of the computer(client) is `172.16.1.6`

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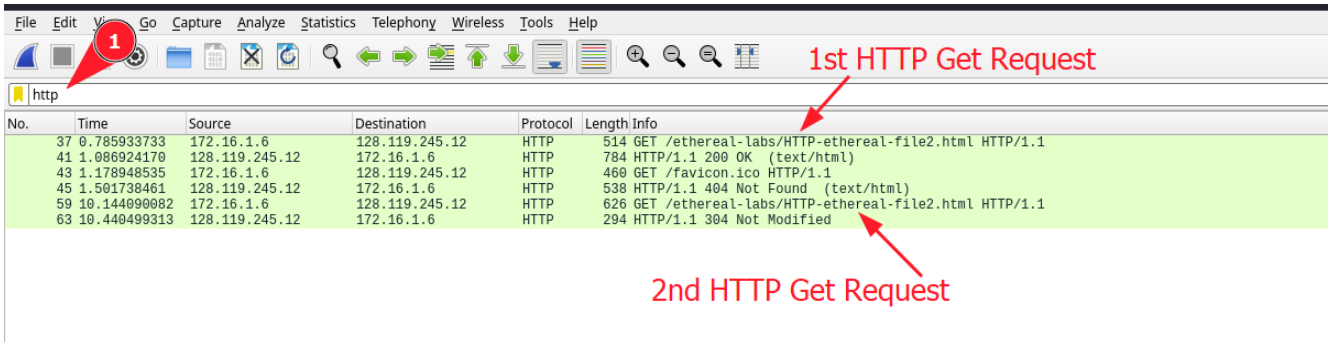
7. Click the following URL Check if this file was modified. If yes? Then mention the date and time. Hint: IF-MODIFIED-SINCE

Packet Capturing procedure:

1. Start the browser and clear the browser's cache.
2. Start capturing the packets using wireshark.
3. Visit the URL "http://gaia.cs.umass.edu/ethereal-labs/HTTP-ethereal-file2.html" using browser.

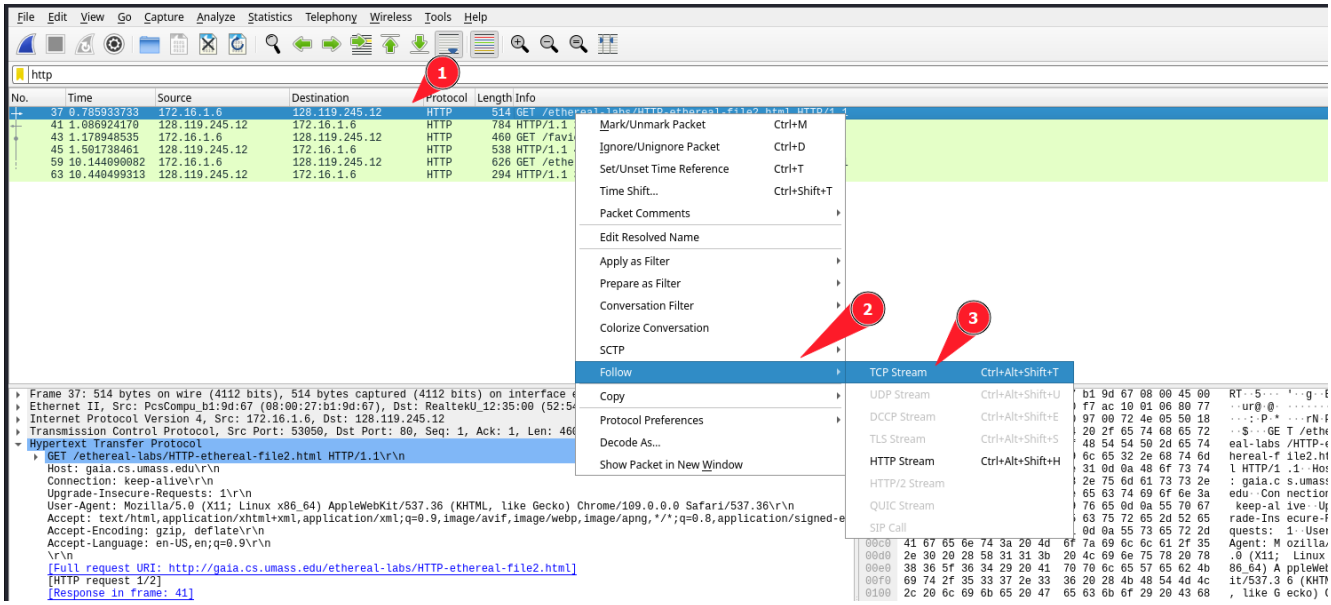


4. Revisit the URL quickly.
5. Stop capturing packets and apply http display filter

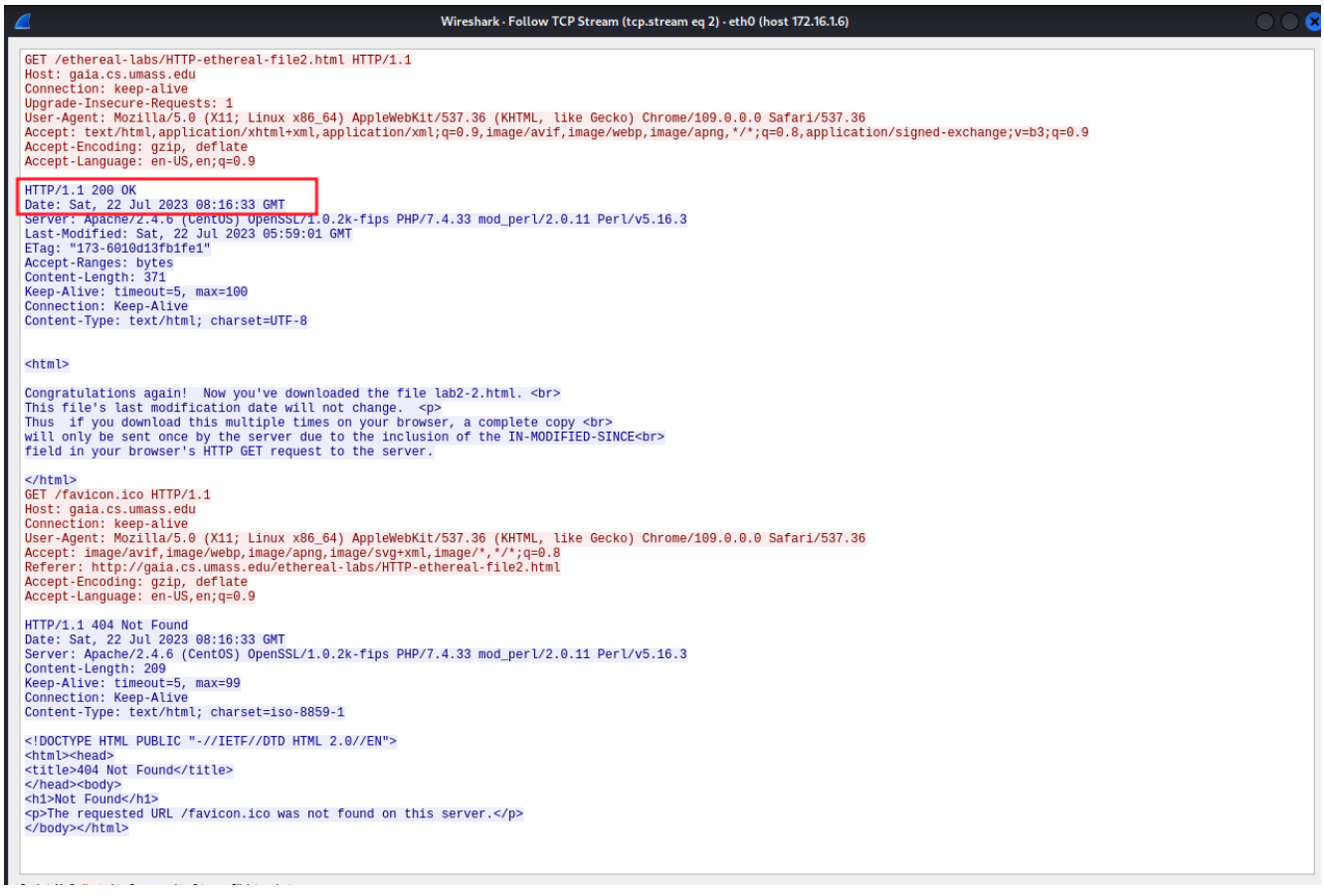


6. HTTP packet analysis: Content of First HTTP Get Request-Response

Right Click on the 1st GET request and select follow -> TCP stream

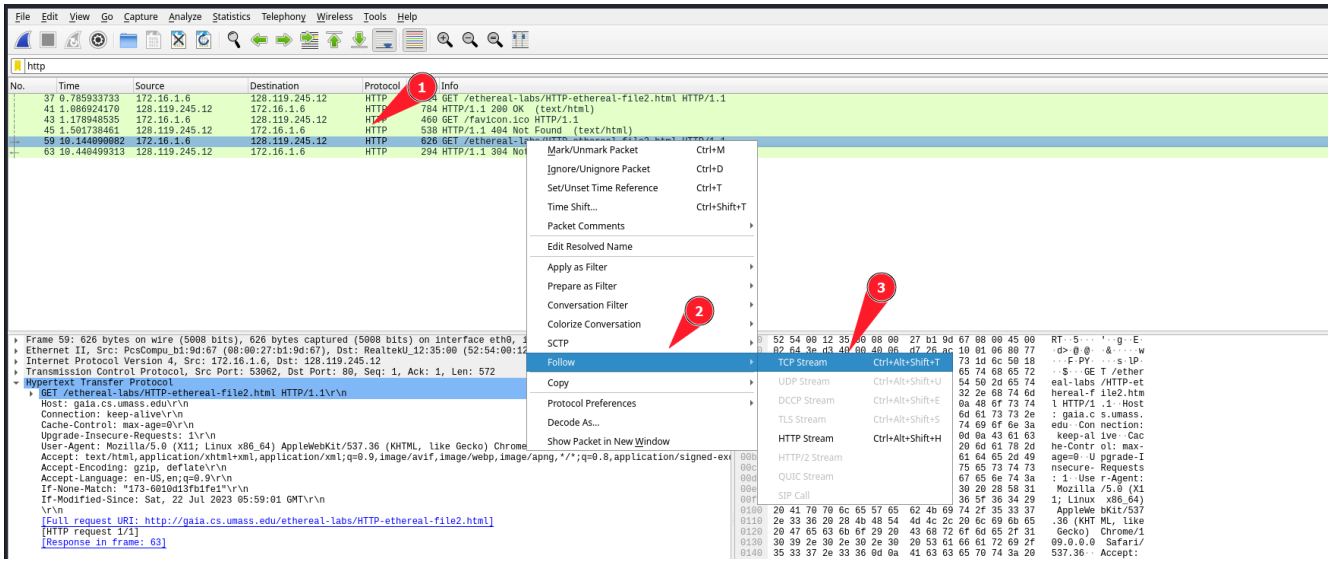


It is observed that on requesting the web-page, server returned the 200 OK status code with the required web-page content.

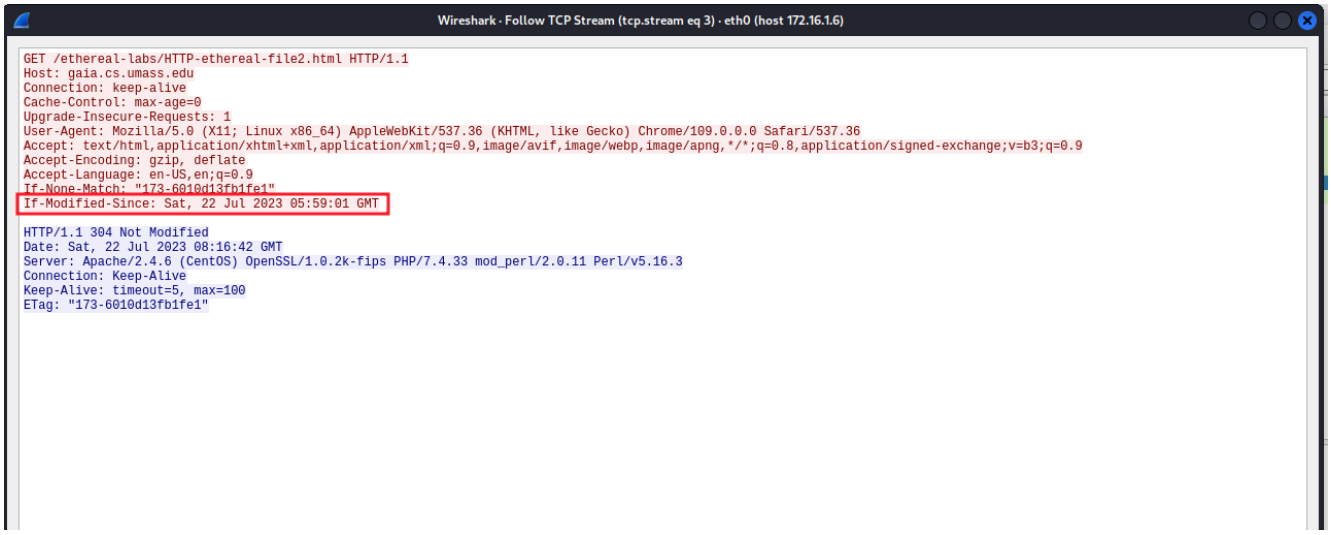


7. HTTP packet analysis: Content of Second HTTP Get Request

Right Click on the 2nd GET request and select follow -> TCP stream



Since the web-page is already cached due to the 1st HTTP request-response. When the 2nd HTTP request is made, server returned the *304 Not Modified* status code.



Date and time of modified file can be obtained by *If-Modified-Since* request header which makes the request conditional. It is based on the condition whether resource is modified after the give date or not. If the resource is modified after the give date then server sends the 200 OK status code, if not then it sends the 304 Not Modified status code as seen in the above picture.

```
If-Modified-Since: Sat, 22 Jul 2023 05:59:01 GMT
```

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8. Briefly explain the function of DNS? Locate the DNS query and response messages. Are they sent over UDP or TCP? Provide a screenshot of Wireshark that includes the Source Port and Destination port for the DNS queried message.

Function of DNS

If user want to visit the facebook, user type "https://www.facebook.com/" in the address bar of the browser. It is also possible to visit the facebook by it's IP address. Thus, host can be identify by two ways : by a hostname and by an IP address. For humans it is easy to use hostname becuase it is easy to remember whereas routers uses IP addresses.

DNS main function is to translate the human-readable domain names, such as www.facebook.com, into machine-readable IP addresses, such as 31.13.79.35.

Let's see how the translation happens. Let's say Alice heard about new social media network say instagram from her friend Bob since he has account on instagram. Now Alice want to create the account on the instagram. She knew the hostname as heard from her friend Bob i.e. www.instagram.com.

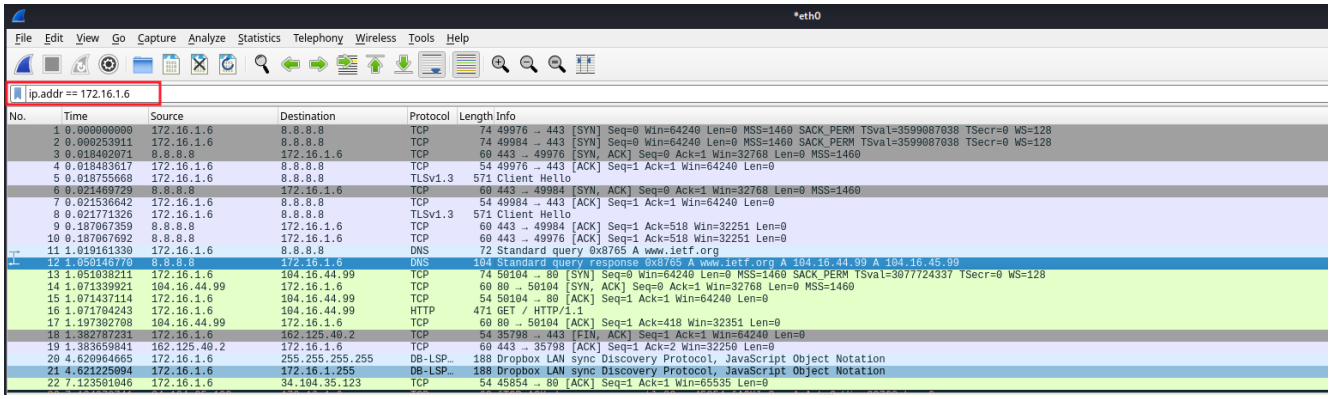
1. Alice opens the browser and entered the URL www.instagram.com in the address bar. Now browser check its own cache whether it has required address or not. If it has then it fetches the page from the instagram server which hosts the instagram. In case it is not present then it asks for IP address to a DNS recursive resolver via ISP (Internet Service Provider).
2. Similary, DNS recursive resolver check its cache, if address is present it answer the query with the address otherwise it asks the root DNS nameserver.
3. Root DNS nameserver sits on the highest level in the hierarchy of the DNS system. It does not the contains the actual address of the instagram server. But it knows who can tell the address i.e. address of the top-level domain (TLD) which include .com in our case. Root DNS tell the DNS resolver about the TLD.
4. Now DNS resolver ask the address of www.instagram.com to TLD nameserver whether it has address or not. TLD nameserver does not contains the IP address but it know who can tell the address i.e. actual DNS server for instagram.com and return the IP address of DNS server to the DNS resolver.
5. Now DNS resolver ask for the IP address to instagram's DNS nameserver (instagram.com)
6. Instagram's DNS nameserver returns the target IP address of instagram to DNS resolver.
7. Once the IP address is resolved by the DNS resolver it tells the browser about the resolved address.
8. Once the browser knows the IP address it sends the HTTPS request to fetch web-page served by the instagram server. Then server response the requested page which is displayed in the Alice browser.

DNS query and response messages

Packet Capturing procedure:

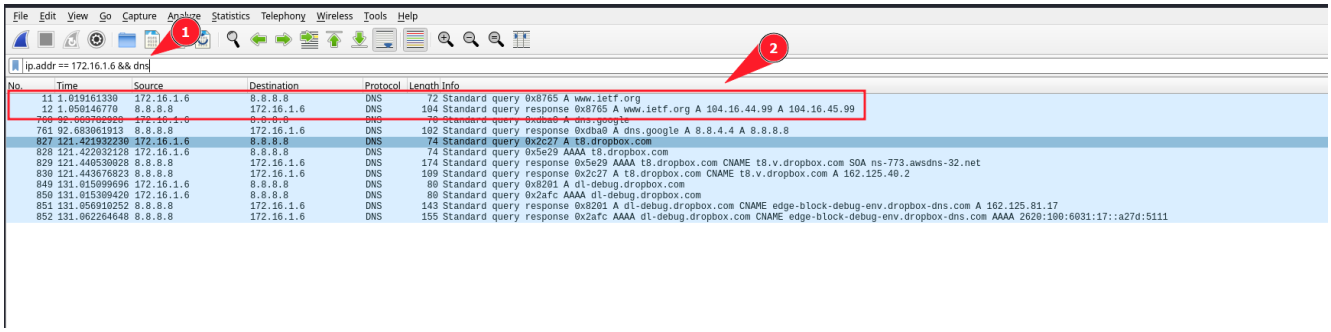
1. Use ipconfig to empty the DNS cache in your host. (for Windows based machines)
2. Open browser and empty browser cache.
3. Open Wireshark and enter "ip.addr == 172.16.1.6" into the display filter. This filter removes all packets that neither originate nor are destined to host.
4. Start packet capture in Wireshark.
5. With browser, visit the Web page: http://www.ietf.org

6. Stop packet capture.

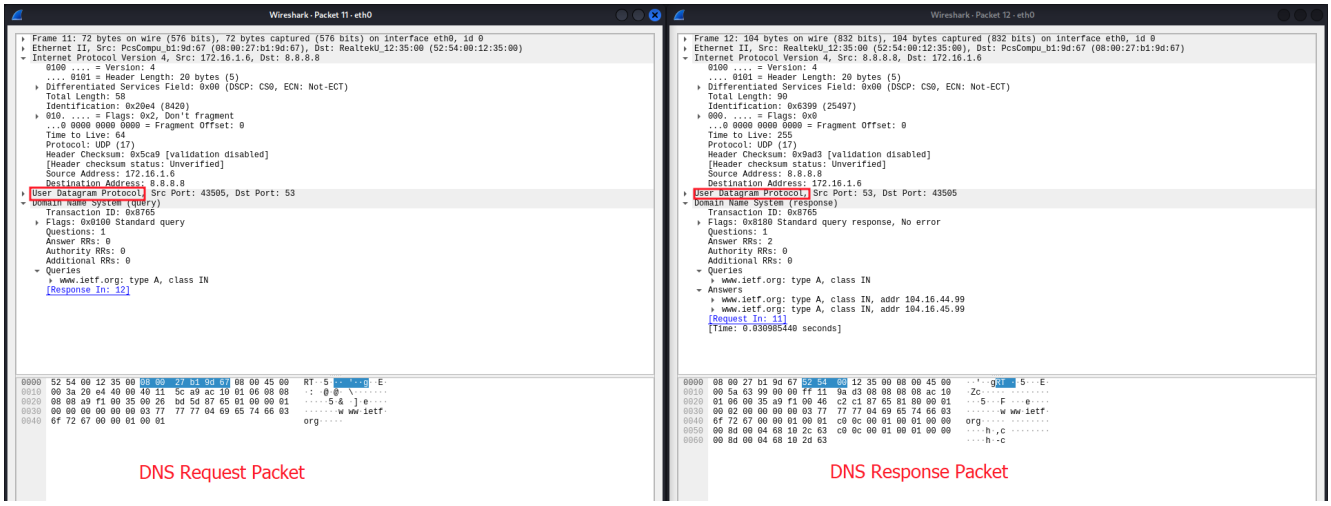


7. DNS query and response packets

   Apply "ip.addr == 172.16.1.6 && dns" as display filter. Packet 11 and packet 12 are DNS query and response packets as shown in below picture.



   Details of DNS request and response packets



> **DNS query packets are sent over UDP**

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9. What are Authoritative and Non-Authoritative name servers? Demonstrate one of them using WireShark.

Authoritative name servers

These are the first source of truth for a specific domain's DNS records. Because When a domain is registered, the domain owner specifies the authoritative DNS servers responsible for managing the domain's DNS records. They contain the updated and original DNS records for a domain and are responsible for providing accurate information about that domain's IP addresses, mail servers, and other associated resources.

For example, if we want to find the IP address of "www.facebook.com," then corresponding DNS resolver will contact the Non-authorative name server and if it does not contains the DNS entry for www.facebook.com then it will fetch it from authoritative name servers for the "facebook.com" domain to get the correct information.

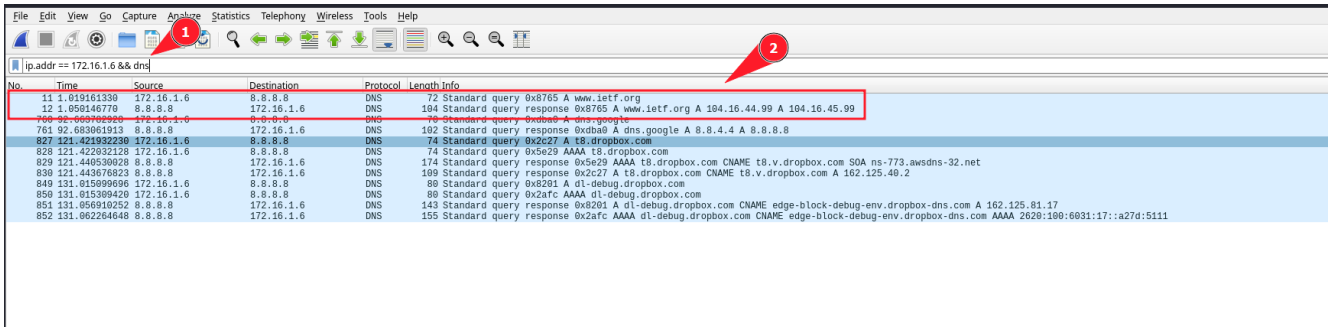
Non-authoritative name servers

These servers are DNS servers that do not host the official DNS records for a domain but cache the information obtained from authoritative name servers. These servers act as intermediaries between end-users and the authoritative name servers. When a user's device or DNS resolver makes a DNS query, non-authoritative name servers are typically the first ones to respond.

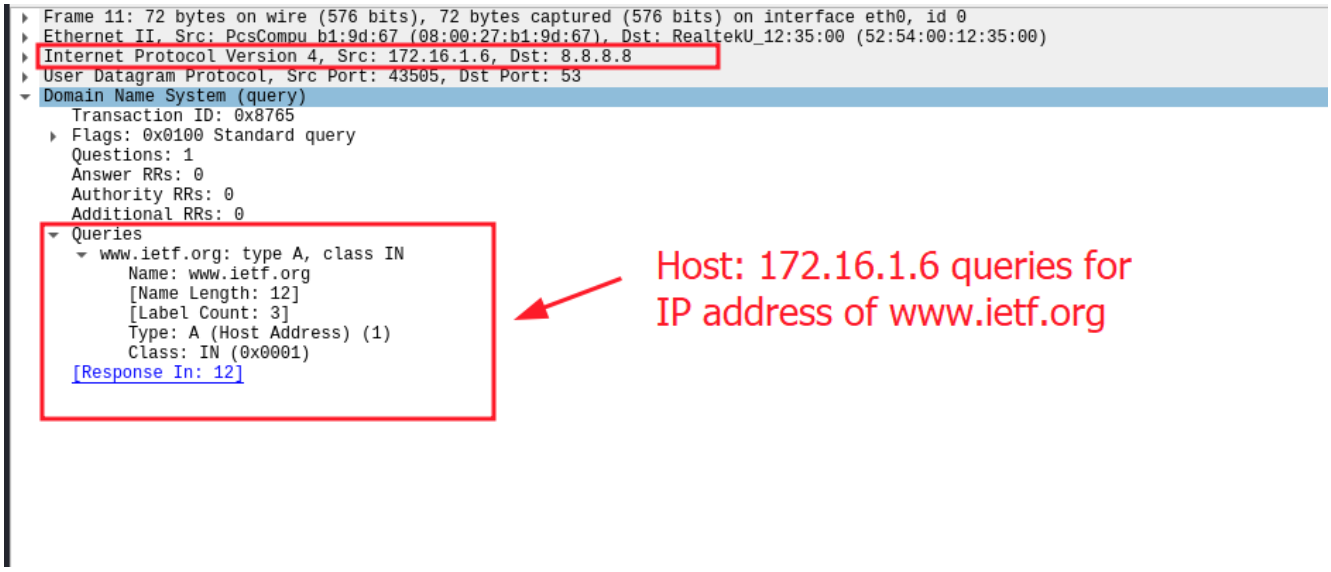
For example, Let say user has already visited the www.facebook.com once and now user wants to visit it again. In this case, DNS resolver contacts the Non-authoritative name server. Since DNS entry is cached non-authoritative name server returns the corresponding entry to resolver.

Packet Capturing procedure:

1. Use ipconfig to empty the DNS cache in your host. (for Windows based machines)
2. Open browser and empty browser cache.
3. Open Wireshark and enter "ip.addr == 172.16.1.6" into the display filter. This filter removes all packets that neither originate nor are destined to host.
4. Start packet capture in Wireshark.
5. With browser, visit the Web page: http://www.ietf.org
6. Stop packet capture.
7. Apply "ip.addr == 172.16.1.6 && dns" as display filter.



We can see that packet 11 and packet 12 are DNS request and response packets. Here host (source: 172.16.1.6) send the dns request packets to non-authoritative name server (destination: 8.8.8.8). 8.8.8.8 is an IP address that corresponds to one of Google's public DNS server. This are popular DNS server due to fast, reliable, and well-maintained query resolution.
Here host (172.16.1.6) ask for the IP address of www.ietf.org using dns query packet (packet 11).



Non-authoritative name server (8.8.8.8) responds with IP address of www.ietf.org



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10. What are the different types of DNS records? Using Wireshark examine the DNS query message and write the "TYPE" of the DNS record. Does the query message contain any "answers"?

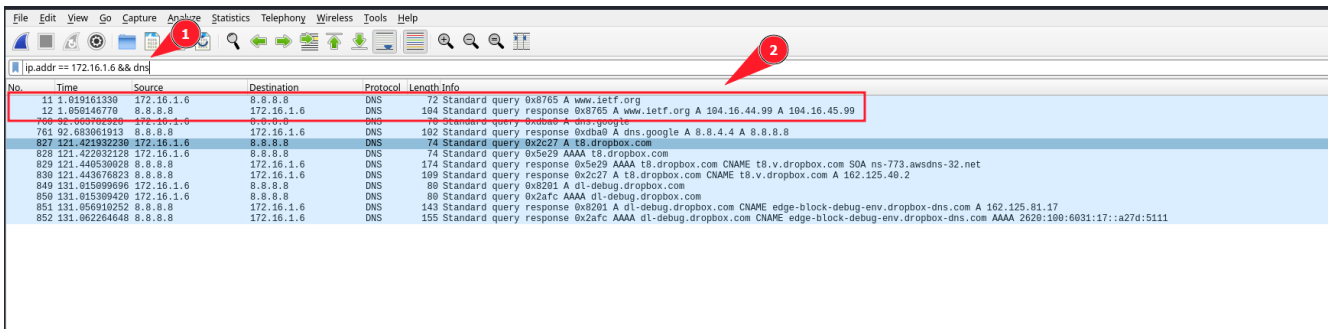
Some of the most commonly used DNS record types are as follows:

1. A Record: It is used to point a domain name to the IP address of the server hosting the website or service. It contains IPv4 addresses.
2. AAAA Record: It contains the IPv6 addresses for a domain.
3. CNAME Record: This record creates an alias for one domain name to another i.e. it forwards one domain or subdomain to another domain.
4. MX Record: This record is used to specify the mail servers responsible for receiving emails for a domain. It points to the domain name of the email server.

5. TXT Record: This record allows domain administrators to add arbitrary text to their DNS entries. These records are frequently used for email security.
6. NS Record: This record indicates the authoritative name servers for a domain. It specifies which DNS servers are responsible for handling queries for a specific domain.
7. SOA Record: This record provides administrative information about the zone, such as the primary name server, contact email, zone serial number, etc.
8. PTR Record: This record is used for reverse DNS lookups. It maps an IP address to a domain name, which is helpful for verifying the ownership of an IP address.
9. SRV Record (Service Record): This record is used to specify the port of services, such as VOIP, FTP, or LDAP.

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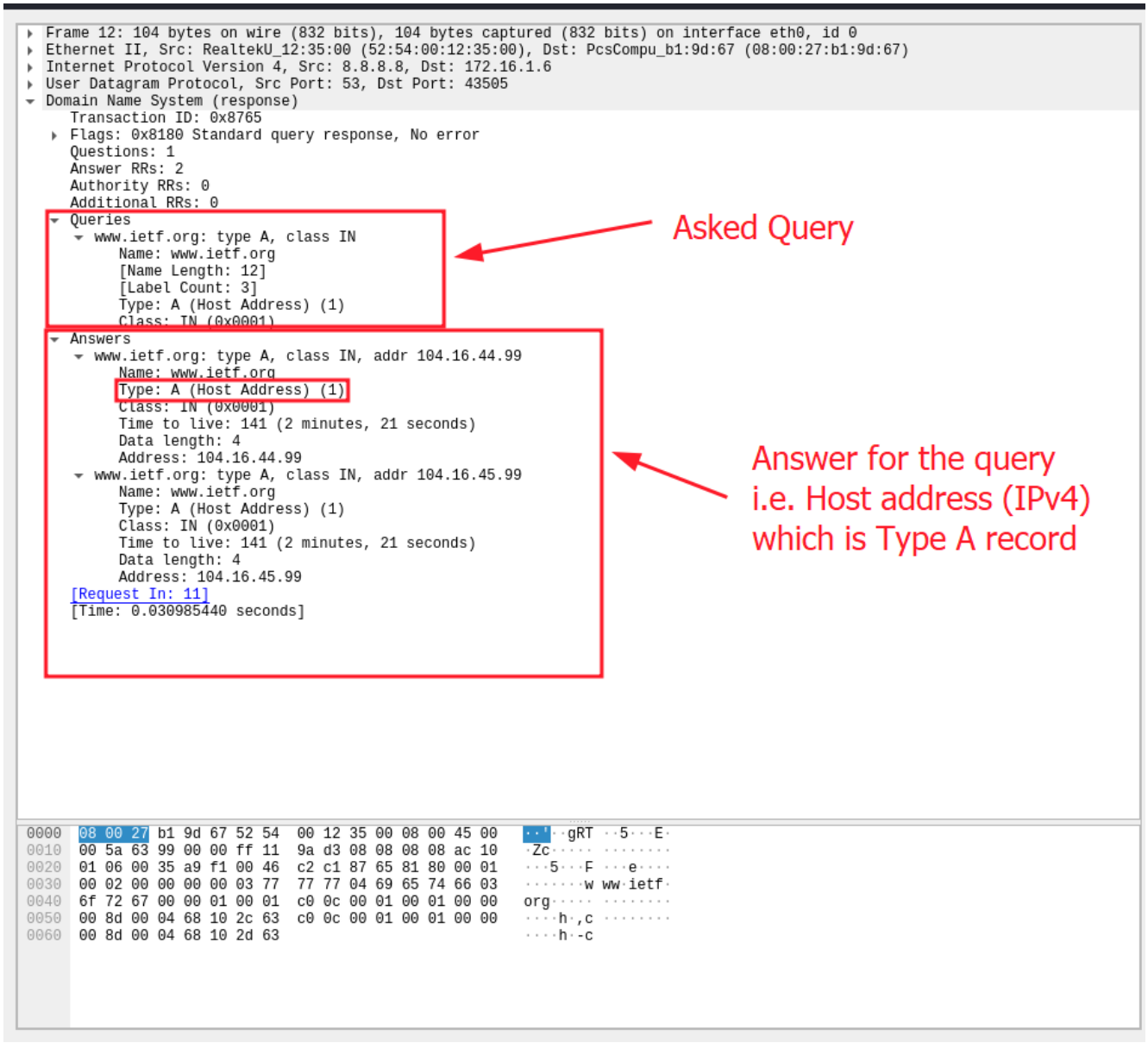


8. Get DNS query response packet

   Click on the query response packet (packet 12). We can observe that query message contains asked query and answer provided by the name server server.

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   The "Type" of the DNS record is of A Type.
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   Query messages also contains **answer** section which provides the Host (IPv4) address of the "www.ietf.org"


Frame 12: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface eth0, id 0
Ethernet II, Src: RealtekU_12:35:00 (52:54:00:12:35:00), Dst: PcsCompu_b1:9d:67 (08:00:27:b1:9d:67)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 172.16.1.6
User Datagram Protocol, Src Port: 53, Dst Port: 43505
Domain Name System (response)
Transaction ID: 0x8765
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 0
Queries
www.ietf.org: type A, class IN
Name: www.ietf.org
[Name Length: 12]
[Label Count: 3]
Type: A (Host Address) (1)
Class: IN (0x0001)
Asked Query
Answers
www.ietf.org: type A, class IN, addr 104.16.44.99
Name: www.ietf.org
Type: A (Host Address) (1)
Class: IN (0x0001)
Time to live: 141 (2 minutes, 21 seconds)
Data length: 4
Address: 104.16.44.99
www.ietf.org: type A, class IN, addr 104.16.45.99
Name: www.ietf.org
Type: A (Host Address) (1)
Class: IN (0x0001)
Time to live: 141 (2 minutes, 21 seconds)
Data length: 4
Address: 104.16.45.99
[Request In: 11]
[Time: 0.030985440 seconds]
Answer for the query
i.e. Host address (IPv4)
which is Type A record
0000 08 00 27 b1 9d 67 52 54 00 12 35 00 08 00 45 00
0010 00 5a 63 99 00 00 ff 11 9a d3 08 08 08 08 ac 10
0020 01 06 00 35 a9 f1 00 46 c2 c1 87 65 81 80 00 01
0030 00 02 00 00 00 00 03 77 77 77 04 69 65 74 66 03
0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01 00 00
0050 00 8d 00 04 68 10 2c 63 c0 0c 00 01 00 01 00 00
0060 00 8d 00 04 68 10 2d 63